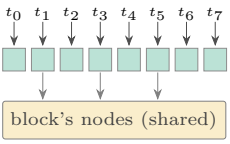
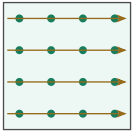
(a) a thread per element



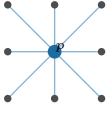
TET4, TET10, HEX8: a warp runs the 32 elements of a store block, so code loads are coalesced; the scatter adds atomically, or through the block's nodes in shared memory.

(b) a thread per point



HEX27 point store: 64 threads per element, one per point, each decoding its point's codes; gradients and outputs by sweeps along the lines of points in shared memory.

(c) a warp per element

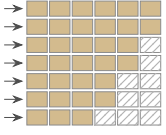


HEX27 element matrix: a lane per node p,

$$y_p = \sum_{q \neq p} r_{pq} K_{pq} (h_q - h_p),$$

from the increment in shared memory, one atomic addition per node.

(d) a thread per row



quantized matrix: SELL-32-1024 slices, one record of codes, column and scale per block; in the full difference form each thread writes its own row, without atomics.